MT8002 Elements of Set Theory
Chapter 6. Cardinals without the Axiom of Choice

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Introduction
Comparing Sizes
Infinite Sets without AC - Countable Sets
Uncountable Sets and Cardinal Arithmetic, without AC
References

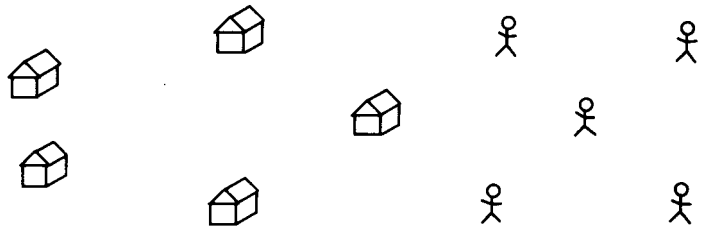
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Introduction

Question

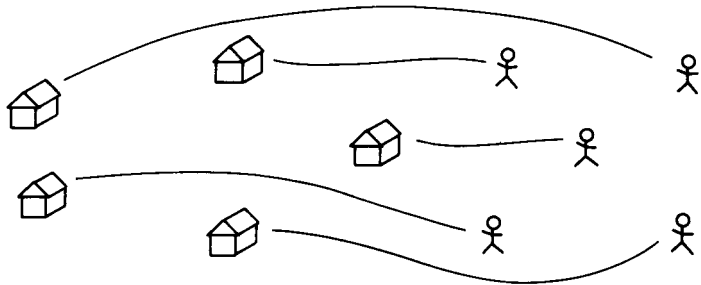
Suppose you only know how to count to three. Are the set of houses and the set of people in the figure the same size?[†]



[†]Figure source: (Enderton 1977, Fig. 30).

Answer

Yes! The set of houses is the same size as the set of people, as shown in the figure.[†]



[†]Figure source (Enderton 1977, Fig. 31).

Introduction

Question

In the previous question, the sets were finite. Can we apply similar reasoning if the sets are infinite?

Introduction

Definitions (p. 127)

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(a) A set X is **finite** if there is a bijective function $f : \mathbf{n} \longrightarrow X$, for some natural number $\mathbf{n}$.

(b) A set X is **infinite** if it is not finite. That is, there is no a bijective function $f : \mathbf{n} \longrightarrow X$, for any natural number $\mathbf{n}$.

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Comparing Sizes

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Comparing Sizes

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Comparing Sizes

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(b) A set A is **less than or equinumerous** with a set B, denoted $A \lesssim B$, if there is a one-to-one function $f : A \longrightarrow B$.

(c) A set A is **dominated** by a set B, denoted $A \prec B$, if $A \lesssim B$ and it is not the case that $A \approx B$.

Comparing Sizes

Notation

$X \not\approx Y := \neg(X \approx Y)$,
$X \not\lesssim Y := \neg(X \lesssim Y)$,
$X \not\prec Y := \neg(X \prec Y)$.

Comparing Sizes

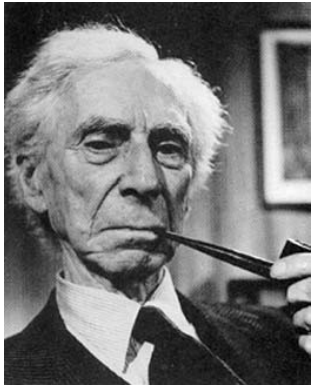
Examples

(1) $\mathbb{N} \precsim \mathbb{Z}$ and $\mathbb{Z} \precsim \mathbb{N}$.

(2) $\mathbb{N} \approx \mathbb{Z}$.

(3) $\mathbb{N} \not\prec \mathbb{Z}$ and $\mathbb{Z} \not\prec \mathbb{N}$.

Comparing Sizes



(1872 – 1970)

"*The possibility that whole and part may have the same number of terms is, it must be confessed, shocking to common-sense.*" *(Russell 1903, p. 358)*

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Infinite Sets without AC - Countable Sets

Definitions (p. 143)

(a) A set A is **countable** if it is finite or equinumerous with $\mathbb{N}$.

Infinite Sets without AC - Countable Sets

Definitions (p. 143)

(a) A set A is **countable** if it is finite or equinumerous with $\mathbb{N}$.

(b) A set A is **countably infinite** if it is equinumerous with $\mathbb{N}$.

Infinite Sets without AC - Countable Sets

Examples

(1) Let $\mathbf{n} \in \mathbb{N}$. The set $\mathbf{n}$ is countable because it has n elements.

(2) The sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countably infinite.

Infinite Sets without AC - Countable Sets

Theorem 6.5 (Cantor's Theorem)

For any set X, $X \prec \mathcal{P}(X)$.

Example

$\mathbb{N} \prec \mathcal{P}(\mathbb{N}) \prec \mathcal{P}(\mathcal{P}(\mathbb{N})) \prec \cdots$.

Infinite Sets without AC - Countable Sets

Definition (p. 148)

A set **uncountable** or **uncountably infinite** is an infinite set which is not countable.

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Uncountable Sets and Cardinal Arithmetic, without AC

Remark

Recall that $\mathbf{2}^X \doteq \{\mathbf{0}, \mathbf{1}\}^X$ is the set of functions from the set X to $\{\mathbf{0}, \mathbf{1}\}$.

Uncountable Sets and Cardinal Arithmetic, without AC

Remark

Recall that $\mathbf{2}^X \doteq \{\mathbf{0}, \mathbf{1}\}^X$ is the set of functions from the set X to $\{\mathbf{0}, \mathbf{1}\}$.

Theorem 6.7

For any set X, $\mathcal{P}(X) \approx \mathbf{2}^X$.

Theorem 6.8

The set $\mathbb{R}$ is uncountable.

Theorem

$\mathbb{R} \approx \mathcal{P}(\mathbb{N}) \approx \mathbf{2}^{\mathbb{N}}$.

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References

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on pp. 5, 6).

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).

Bertrand Russell (1903). The Principles of Mathematics. Cambridge University Press (cit. on p. 16).